

Computer science **Engineer** from higher national school of computer science and **Phd student at Cnam Paris**, specialized in data science and artificial intelligence interested in the field of attack detection in cyber physical system using graph neural networks and Transformers.

SKILLS:

- Communication and teamwork
- Problem solving (C,Java,python)
- Image Processing & Natural Language Processing
- algorithms parallelization with Cuda ,Mpi, Pthreads
- Machine learning & Data Science with python
- Business intelligence (Tableau, Pentaho ETL)
- Back-end developer with Django

Certificates:

- Data base management with SQL
- Frontend fundamentals (pirple-2020)
- Deep Learning & Machine Learning (Coursera)
- CCNA: Introduction to Networks by Cisco – 2021
- Machine learning and Data engineering (Kaggle)
- JavaScript & JQuery Tutorial (SoloLearn)

FORMATIONS

2016 - 2018
Batna, Algeria

Baccalaureate exam (17.03/20) with honors
mixed high school in n'gaous

2018- 2023
Alger, Algeria

ESI School.
higher national school of computer science
Engineer in intelligent systems and data science

2023- 2027
Paris,France.

Phd Student: CNAM Paris.
GNN-based Attack detection in cyber physical and IoT systems.
Diploma: PhD in Computer Science.

EXPERIENCES

Oct. 2022 – Jan. 2023
Algeria, Alger

Fire Detection
A deep learning model for forest fire detection

- A 3-month internship in deep learning to be able to detect forest fires in real time using several learning concepts and algorithms like yolo and RCNN.

Oct. 2022 – Jua. 2023
Algeria, Alger

Word Spotting / LCSI-Lab
A deep learning model for the topic of word spotting

- 9 months, keyword detection is a technique of finding all occurrences of a specific word or phrase in a given data source, such as an audio recording or scanned document image, without fully recognizing the word or phrase.

Mar. 2022 – Jua. 2022
Algeria, Alger

Optimization Project

the permutation flow shop optimization

- use heuristics and meta heuristics and even hyper heuristics to optimize this problem and propose an approach based on reinforcement learning (Python) in order to choose the best hyper heuristics that maximizes the objective function.

Mar. 2022 – Jul. 2022
Algeria, Alger

Community detection

Project in data science

- Project in which I collaborated with a team of six members in the development of a solution that can outperform the existing solutions in the field of community detection using a hybridization of machine learning and metaheuristics (Python)

Academic teaching experience:

- ✓ Professional Diploma in Web Integration Assistance – **USAL33 : HTML/CSS (More than 60h).**
- ✓ Professional Diploma in Web Integration Assistance – **Networking (More than 20h).**
- ✓ Supervision of a master's student on IoT attack detection using machine learning **2024-2025 (CNAM)**

Research Papers:

1. [Leveraging_Graph_Neural_Networks_for_Attack_Detection_in_IoT_Systems 2025, ARES conference.](#)
2. Transformer-Based Multi-Class Intrusion Detection for IoT and Network Traffic Analysis
(Conference Under Evaluation)
3. Disentangling Spatial and Temporal Context for APT Stage Detection: A Graph Learning Study on Unraveled
(Journal Under Review)
4. Endpoint-Relation-Aware Line Graph Neural Networks for Multiclass IoT Intrusion Detection, (Journal under preparation)

LANGUAGES:

- ✓ FRENCH: Fluent
- ✓ ENGLISH: Fluent